INF 201 – Advanced Programming

Deliverable 2 – AI declaration

What I have used AI for

I pasted in the assignment text so that the AI could give me relevant and constructive feedback. I have solved the tasks myself and written all of the code/large parts of the code myself. I used Claude as support in the following ways:

- ***Outline:*** I asked for suggestions on how the task could be split into functions, without getting finished code.
- ***Hints when I was stuck:*** I asked specific questions about individual methods and concepts (see the list below) and used the explanations to write the code myself.
- ***Naming:*** I asked for suggestions for names of functions and variables
- ***Wording of docstrings:*** I got help wording the docstrings for the different functions in my script.
- ***Error checking and PEP8:*** I asked the AI to look over the code for errors, typos and PEP8 violations. The changes were made manually by myself.
- ***Generating a script:*** I got help from the AI to generate a script/bullet list for talking about and presenting my code in the best possible way.

I used AI to sort my questions chronologically and to word the AI declaration and the summary. Suggested changes made by the AI (Claude) were assessed and manually changed by me without copying and pasting.
The questions below are reproduced approximately and have been sorted so that they appear in the same order as in the code in the script.

Task 1: Reading files and regular expressions

The assignment text and outline

1. **The assignment text**
   "Here is the assignment text for deliverable 2: (paste). I am going to solve tasks 1–3, and my partner will use my functions in tasks 4 and 5. Can you read the assignment and tell me what I should be aware of, without telling me how to solve it?"
2. **The data generation script**
   "We were given a script called generate_data_for_d2.py. Do I need it to solve the assignment? What do I need to change for it to work on my Mac?"
3. **Splitting into functions**
   "Task 1 asks for a function that reads files and a function that parses log lines. What should each function be responsible for, and what should they be called? Don't write the code."

Reading files with different character encodings

4. **Two character encodings**
   “The task says that the files are either UTF-8 or ISO-8859-1. What is the difference between them, and is ISO-8859-1 the same as Latin-1?”
5. **Finding the encoding without hard-coding**
   “How can the function find out which character encoding a file has without me writing it in for each file? In week 41 I used read_bytes(), decode() and try/except. Can I use the same technique here?”
6. **The order of the attempts**
   “Does it matter whether I try UTF-8 or ISO-8859-1 first? What happens if I try Latin-1 first?”
7. **Type hints and return value**
   “The function should be able to take both a Path and a normal string, and return a tuple with two strings. How do I write type hints for that?”

Regular expressions

8. **Getting started with the pattern**
   “I am going to make a regex pattern that matches this line: [2026-10-15 08:30:00] [POS_01] item=Latte price=4.50 EUR (SUCCESS). How can I build the pattern piece by piece, instead of writing everything at once?”
9. **Characters that need a \ in front**
   “Why do I have to write [, (and . in the pattern? What do [ ], ( ) and . mean in regex if I don’t have a \ in front?”
10. **Digits and letters**
    “What is the difference between \d, \w and [A-Z]? How do I say ‘exactly four digits’ or ‘three capital letters’?”
11. **Decimal numbers**
    “How do I make a pattern that matches a decimal number like 4.50, but not just a full stop on its own?”
12. **One of several values**
    “The status can only be SUCCESS, REFUND or CANCELLED. How do I write ‘one of these’ in regex?
13. **Named groups**
    “I want to extract six fields from each line and get them as a dictionary. How do I use named groups (?P<name>…) and .groupdict()?”

Parsing the log text

Going through the lines
“How do I go through the text line by line? Why should I use strip() on each line before I match the pattern?”

Skipping noise
“What does .match() return when a line doesn’t fit the pattern? How do I skip those lines with an if and continue?”

Price as a number
"The task says that price should be a float, but regex gives me text. How do I convert the price before I add the transaction to the list?"

Error checking and PEP8

17. **SyntaxError on import**
    "I get SyntaxError: invalid syntax on the line Import re. What is wrong?"
18. **AttributeError**
    "I get AttributeError: 'str' object has no attribute 'splittlines'. What does the error message mean, and how do I find the error?"
19. **IndexError and 0 transactions**
    "My test says that the file has 90 lines, but parse_log finds 0 valid transactions, and I get IndexError: list index out of range. How do I find out whether the error is in the regex pattern or in the function? Don't fix it for me, just give me a way to debug it."
20. **Testing the functions**
    "How can I use assert to test that parse_log ignores noise lines and only finds the valid line?"
21. **Results and discussion**
    "What should I include in 'Results and discussion' for Task 1? Why were all the files read as UTF-8 even though some are written with ISO-8859-1? Give me keywords, not finished text."

Task 2: Recursive utility functions

The assignment text and outline

1. **The assignment text**
   "Here is Task 2: (paste Task 2). Can you read it and tell me what I should be aware of, without telling me how to solve it?"
2. **Splitting into functions**
   "Task 2 asks for one function that finds log files and one that finds active registers. Do I need a separate function to read the JSON file? Suggest names, but don't write the code."

Finding log files recursively

4. **Going through a folder**
   "How do I use Path.iterdir() to go through everything in a folder? How do I check whether something is a folder or a file?"
5. **Calling the function on itself**
   "When the function finds a subfolder, it should continue searching inside it. How do I make the function call itself and add the result to the list?
6. **Base case**
   "What is the base case in a function that searches through folders? How does the function know when to stop?"

7. **Only .log files**
   "How do I check that a file ends with .log without using glob? Why is it a good idea to have the file extension as an argument with a default value?"
8. **Fixed order**
   "The order the files come in from iterdir() can vary. How can I make sure they always come in the same order?"

Finding active registers recursively

9. **Reading the JSON file**
   "I have used json.dump to write JSON in week 40. How do I read a JSON file into a Python dictionary?"
10. **The structure of the config file**
    "store_config.json has dictionaries inside lists inside dictionaries. How can a function go through such a structure when it doesn't know how many levels there are?"
11. **Finding an active register**
    "How do I check whether a dictionary is a register that is active? Why should I check that both "id" and "active" exist before I use them?"
12. **Base case for nested data**
    "What should the function return if data is neither a dictionary nor a list, for example a text like "Campus Coffee"?"

Error checking and PEP8

13. **Final review**
    "Here is my code for Task 2: (paste). Can you check that it meets all the requirements in the assignment, and look for PEP8 violations, typos and docstrings that don't match the parameters? Don't rewrite it, just point out which lines I should look at more closely."
14. **Results and discussion**
    "What should I include in 'Results and discussion' for Task 2? What is the advantage of using recursion here? Give me keywords, not finished text."

Task 3: Cleaning and analysing data with NumPy

The assignment text and outline

1. **The assignment text**
   "Here is Task 3: (paste Task 3). Can you read it and tell me what I should be aware of, without telling me how to solve it?"

Removing outliers

2. **From list to NumPy array**
   "The function should convert the prices into a 1D NumPy array of floats. How do I do that with np.array?

3. **Mean, standard deviation and median in NumPy**
   "In deliverable 1 we used df["temperature"].mean() and .median() in pandas. Which NumPy functions do the same for an array? What does np.std do, and why do I need np.abs in the formula?"
4. **Calculating on the whole array at once**
   "How do I calculate the distance from the mean for all prices at once, without a for loop?
5. **Comparing an array with a number**
   "What happens when I write distance > 3 * std_price, where distance is an array? Why do I get an array of True and False back, and not just a single True or False?"
6. **Changing only some elements**
   "How do I replace only the outliers with the median and leave the other prices as they are? What does cleaned_prices[is_outlier] = median_valid do?"
7. **Loop or vectorisation?**
   "I could solve this with a for loop and if. Why is it better to use NumPy operations on the whole array?

Refund ratio

8. **What does 1e-6 mean?**
   "The task says that $\varepsilon = 10^{-6}$ should be added to the denominator. How do I write 10^{-6} in Python? What does the notation 1e-6 mean, and why is it a good idea to store it as a constant in capital letters?"

Testing

9. **Testing on the real data**
   "How can I use the functions from tasks 1 and 2 to collect all the prices from the log files, so that I can test the outlier function on them? How do I show that the outliers are gone?"

Error checking and PEP8

10. **NameError in the loop**
    "I get NameError: name 'log_file' is not defined in the test cell. What does the error mean, and how do I find out where it comes from?"
11. **ValueError about broadcasting**
    "I get ValueError: operands could not be broadcast together with shapes (2,0,5) (3,) in compute_refund_ratio. What does the error mean? What is the difference between np.array and np.ndarray?"
12. **Final review**
    "Here is my code for tasks 1–3: (paste). Can you check that it meets all the requirements, and look for PEP8 violations, typos and docstrings that don't match the parameters? Don't rewrite it, just point out what I should look at more closely."
13. **Results and discussion**
    "What should I include in 'Results and discussion' for Task 3? Give me keywords, not finished text."

Generative AI Use – Tasks 4 and 5

I used ChatGPT as a learning and troubleshooting assistant while completing Tasks 4 and 5 of INF201 Deliverable 2.

Task 4 – Pandas Data Analysis

I asked ChatGPT questions about how to convert transaction data into a Pandas DataFrame, calculate hourly revenue, create a pivot table, and calculate refund ratios using the function from Task 3.

ChatGPT helped me understand the code, review the results, and improve the function docstrings and Results and Discussion section.

Task 5 – Full Workflow

I asked ChatGPT how to combine the functions from Tasks 1–4 into a complete workflow, apply price outlier cleaning, correct NOK prices, and generate the required reports.

ChatGPT provided explanations and code suggestions, helped me troubleshoot and verify the results, and assisted with code comments and documentation.

My Contribution

I worked through the tasks step by step, executed the code in VS Code, checked the generated results, and made the final changes. I used ChatGPT for explanations, code suggestions, and writing assistance, but reviewed and tested the final solution myself.